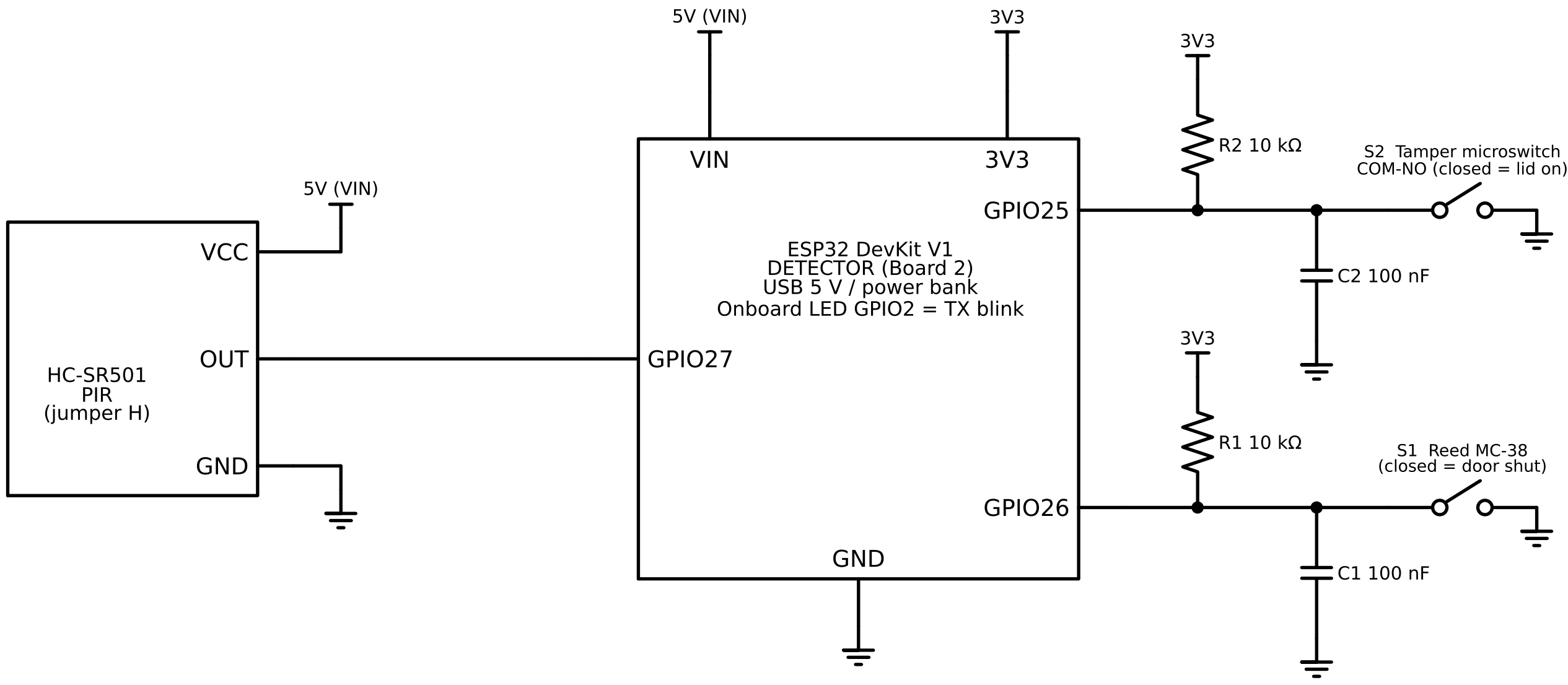

HomeGuard - DETECTOR schematic (25ES604 ESD mini-project) rev 1 2026-11-05

Fail-safe inputs: switch to GND + pull-up, so a cut wire reads OPEN (alarm side). Internal pull-up also enabled.

RC filter: 10 kΩ x 100 nF = 1 ms; firmware debounce 50 ms. PIR needs 5 V supply; OUT is 3.3 V logic (safe for GPIO).